PRE-ALGEBRA (8TH) · UNIT 1: REAL NUMBERS AND EXPONENTS

Space Mission Planner

Use scientific notation, exponents, and roots to plan a crewed mission to Mars.

Honors

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

NASA has asked your class to help plan the first crewed trip to Mars. The distances are enormous, the radio messages take minutes to arrive, and every module of the habitat has to be sized exactly.

You will use scientific notation, exponent rules, and square and cube roots to plan the trip and design the habitat.



Mission Data

Earth to Moon: 3.84×10^5 km

Earth to Mars (closest): 5.46×10^7 km

Speeds

Rocket: 4×10^4 km per hour

Light: 3×10^5 km per second

Learning Goals

- I can solve multi-step problems with scientific notation and unit conversions.
- I can simplify expressions with integer exponents and explain the rules.
- I can reason about irrational numbers and approximate them.

Project Timeline

When	What you will do
Day 1	Beyond Mars
Day 2	Speed & Time
Day 3	Habitat Engineering
Final	Mission Plan + reflection

Part 1: Beyond Mars

Day 1 · A light-year is about 9.46×10^{12} km.

Mission Data Cards

Moon	Mars (closest)	Mars (farthest)
3.84×10^5 km	5.46×10^7 km	4.01×10^8 km
from Earth	from Earth	from Earth

Rocket Speed	Radio Signal	Food
4×10^4 km	3×10^5 km	2.5×10^3 calories
per hour	per second	per astronaut per day

1. Proxima Centauri is 4.24 light-years away. Find the distance in kilometers in scientific notation (three significant digits).

2. How many Earth–Moon distances (3.84×10^5 km) is that? Use scientific notation.

3. A solar film is 0.00000052 m thick. Write this in scientific notation. How many layers make 1 mm?

Part 2: Speed & Time

Day 2 · Use the Mission Data Cards.

4. At 4×10^4 km per hour, how many years would the rocket take to reach Proxima Centauri? (Use 8,760 hours per year.)

5. What speed (km per hour) is needed to reach Mars at its closest in 30 days? Use scientific notation.

6. Mission control sends a question to Mars (closest) and waits for the reply. What is the shortest round-trip wait?

Part 3: Habitat Engineering

Day 3 · Use exponent rules, roots, and reasoning.

7. Simplify $(2^5 \cdot 2^{-2}) \div (2^3)^2$ and evaluate.

8. One cube module has a volume of 1,000 m^3. A storage module has 8 times the volume. Find both edge lengths and the ratio of the edges.

9. Show by squaring that $\sqrt{2}$ is between 1.41 and 1.42. Why does its decimal never end or repeat?

10. The food budget is 1×10^7 calories. Choose a crew size and find the longest mission it supports at 2.5×10^3 calories per person per day.

Final Product: Mission Plan

Your final product must include:

- Interstellar distance and travel time in scientific notation
- Required speed and communication delays
- Exponent simplifications with explanations
- Habitat module dimensions
- Mission proposal with a crew-size and food-budget recommendation

Reflection

R1. Explain why $a^{-n} = \frac{1}{a^n}$ using a pattern of powers.

R2. Why can the product of two irrational numbers be rational? Give an example.

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Scientific notation computations, exponent rules, and roots are used correctly to plan the mission. Goes beyond the minimum with a creative, realistic choice.	Scientific notation computations, exponent rules, and roots are used correctly to plan the mission.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Mission plan shows distances, times, supplies, and habitat sizes with labeled work and units. Polished, organized, and visually engaging.	Mission plan shows distances, times, supplies, and habitat sizes with labeled work and units. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.